

Honors Intro Micro | MiniExam 6 Demo

Question 1

Suppose the demand for X is given by

$$P = 100 - Q$$

with a single seller A producing X at a constant marginal cost of 10. The marginal revenue for A is

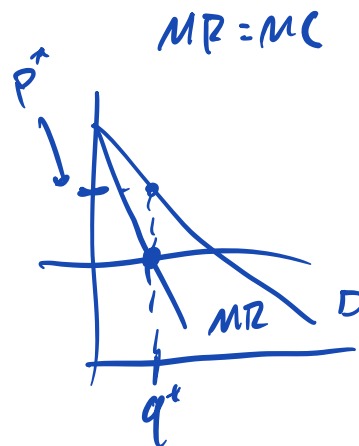
$$MR_A = 100 - 2q_A$$

Prices are in galleons.

Part A. Equilibrium Quantity

What quantity of X should A supply?

$$\begin{aligned} MR &= MC \\ 100 - 2q_A &= 10 \\ 2q &= 90 \Rightarrow q^* = 45 \end{aligned}$$



Part B. Equilibrium Price

What price should A charge?

$$\begin{aligned} P &= 100 - 45 \\ P^* &= 55 \end{aligned}$$

Part C. Profit

What profit will A generate?

$$\begin{aligned}\pi &= TR - TC \\ &= P \cdot Q - MC \cdot Q \\ &= Q(P - MC) \\ &= 45(55 - 10)\end{aligned}$$

$$\pi = 45^2$$

Part D. Post Breakup Quantity

Due to its strangle-hold on the market, the Commerce branch of the Ministry of Magic imposed a heavy-handed breakup of A into two companies, B and C, both with a marginal cost of 10 and with symmetric marginal revenue:

$$MR_B = 100 - 2q_B - q_C$$

$$MR_C = 100 - 2q_C - q_B$$

Find the Nash Equilibrium quantity of X supplied by B.

$$\begin{aligned}MR_B &= MC_B \\ 10 &= 100 - 2q_B - q_C \\ 2q_B &= 90 - q_C\end{aligned}$$
$$\begin{aligned}q_B &= 45 - \frac{1}{2}q_C \\ q_C &= 45 - \frac{1}{2}q_B \\ \frac{3}{4}q_B &= \frac{45}{2}, q_B = \frac{90}{3}\end{aligned}$$

Part E. Post Breakup Market Equilibrium Quantity

What is the market equilibrium quantity in the market? Is quantity less than, equal to, or greater than profit before the breakup?

$$Q^* = q_B + q_C = 60$$

The equilibrium quantity post-breakup is greater.

Part F. Post Breakup Price

What the market price in market after the breakup? Is price less than, equal to, or greater than profit before the breakup?

$$P^* = 100 - 60 = 40$$

The price is lower post-breakup.

Part G. Post Breakup Profit

What the total profit (from both firms) in this market? Is profit less than, equal to, or greater than profit before the breakup?

$$\pi_B = q_B(P - MC) = 30(40 - 10) = 30^2$$

$$\hat{\pi} = 2 \cdot 30^2$$

$$\pi = 45^2$$

$$\pi > \hat{\pi}$$